

SOFTWARE ENGINEERING

Lab 2: Agile Backlog Creation & Sprint Simulation in Jira

OpenAPI Mock Server Generator

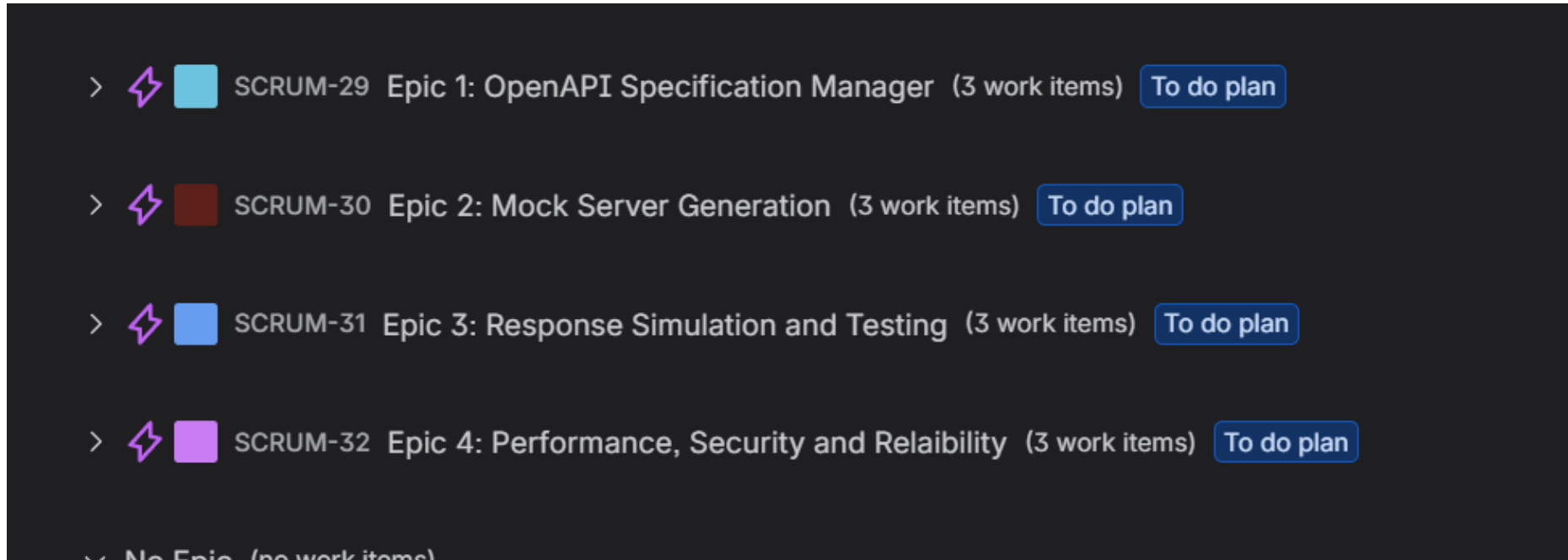
Jira Scrum Backlog and Sprint Report

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Course	UE24CS341A : Software Engineering

Workspace Link - <https://arin-singh-pes1ug24cs079.atlassian.net/jira/software/projects/SCRUM/boards/1/backlog?epics=visible>

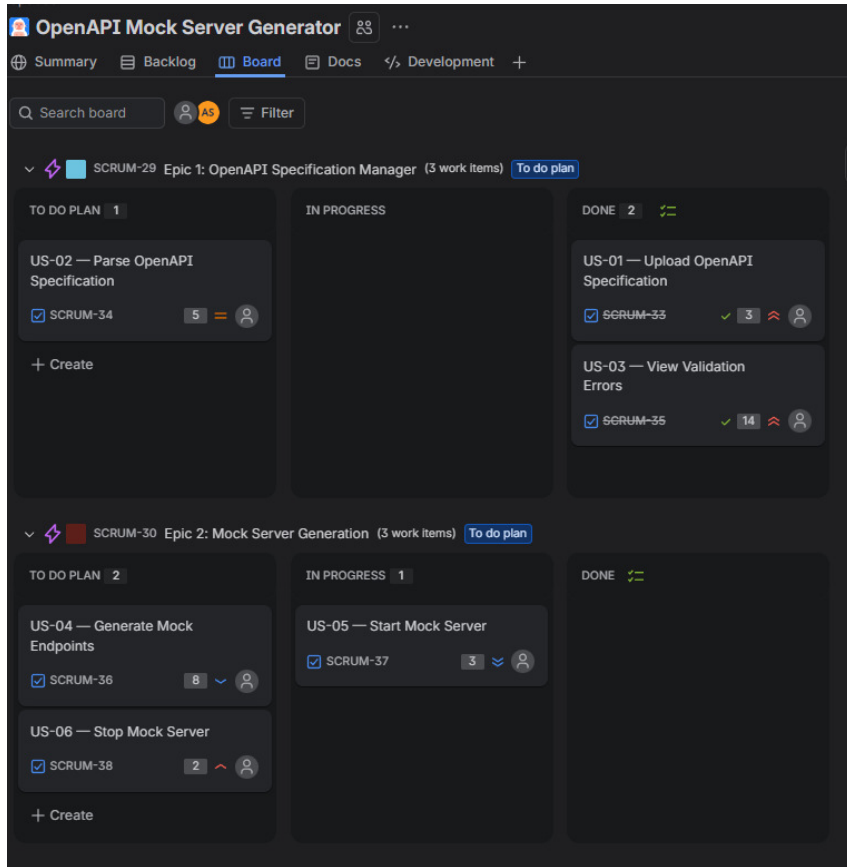
1. Epics and Backlog Structure

The Jira project is organized into four Epics with their related User Stories.



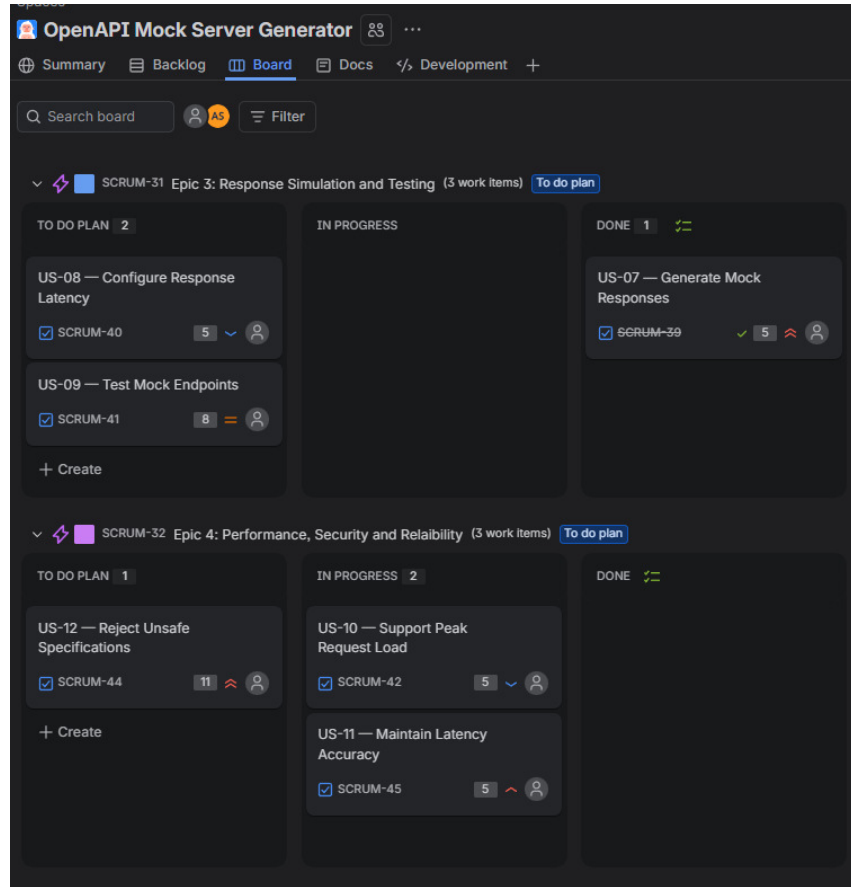
All the Epics Together

Epic Boards with User Stories



The screenshot displays the 'OpenAPI Mock Server Generator' application interface. The 'Board' tab is selected, showing two epic boards. The first epic board, 'Epic 1: OpenAPI Specification Manager' (SCRUM-29), has three columns: 'TO DO PLAN' (1 item), 'IN PROGRESS', and 'DONE' (2 items). The 'TO DO PLAN' column contains 'US-02 — Parse OpenAPI Specification' (SCRUM-34). The 'DONE' column contains 'US-01 — Upload OpenAPI Specification' (SCRUM-33) and 'US-03 — View Validation Errors' (SCRUM-35). The second epic board, 'Epic 2: Mock Server Generation' (SCRUM-30), also has three columns: 'TO DO PLAN' (2 items), 'IN PROGRESS' (1 item), and 'DONE'. The 'TO DO PLAN' column contains 'US-04 — Generate Mock Endpoints' (SCRUM-36) and 'US-06 — Stop Mock Server' (SCRUM-38). The 'IN PROGRESS' column contains 'US-05 — Start Mock Server' (SCRUM-37). Each user story card includes a title, a status, a priority, and an assignee icon.

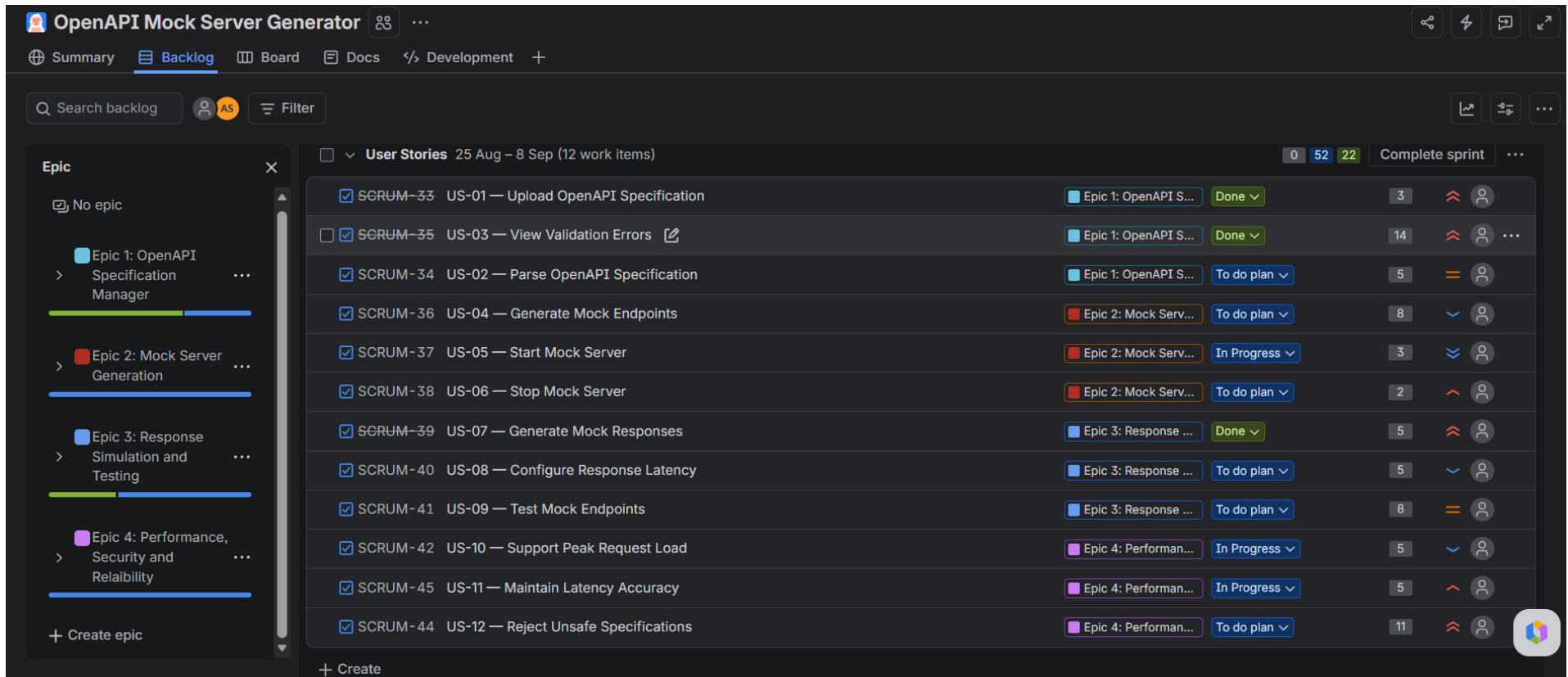
Epic 1 and Epic 2 on the Board with User Stories



The screenshot displays the 'OpenAPI Mock Server Generator' application interface. The 'Board' tab is selected, showing two epic boards. The first epic board, 'Epic 3: Response Simulation and Testing' (SCRUM-31), has three columns: 'TO DO PLAN' (2 items), 'IN PROGRESS', and 'DONE' (1 item). The 'TO DO PLAN' column contains 'US-08 — Configure Response Latency' (SCRUM-40) and 'US-09 — Test Mock Endpoints' (SCRUM-41). The 'DONE' column contains 'US-07 — Generate Mock Responses' (SCRUM-39). The second epic board, 'Epic 4: Performance, Security and Reliability' (SCRUM-32), also has three columns: 'TO DO PLAN' (1 item), 'IN PROGRESS' (2 items), and 'DONE'. The 'TO DO PLAN' column contains 'US-12 — Reject Unsafe Specifications' (SCRUM-44). The 'IN PROGRESS' column contains 'US-10 — Support Peak Request Load' (SCRUM-42) and 'US-11 — Maintain Latency Accuracy' (SCRUM-45). Each user story card includes a title, a status, a priority, and an assignee icon.

Epic 3 and Epic 4 on the Board with User Stories

Epic and User Story Overview



The screenshot displays the Jira interface for the 'OpenAPI Mock Server Generator' project. The 'Backlog' tab is active, showing a list of User Stories grouped under four Epics. The left sidebar shows the Epic hierarchy with progress bars.

Epic	User Story ID	User Story Description	Status	Assignee	Estimate
Epic 1: OpenAPI Specification Manager	SCRUM-33	US-01 — Upload OpenAPI Specification	Done	AS	3
	SCRUM-35	US-03 — View Validation Errors	Done	AS	14
Epic 2: Mock Server Generation	SCRUM-34	US-02 — Parse OpenAPI Specification	To do plan	AS	5
	SCRUM-36	US-04 — Generate Mock Endpoints	To do plan	AS	8
	SCRUM-37	US-05 — Start Mock Server	In Progress	AS	3
Epic 3: Response Simulation and Testing	SCRUM-38	US-06 — Stop Mock Server	To do plan	AS	2
	SCRUM-39	US-07 — Generate Mock Responses	Done	AS	5
	SCRUM-40	US-08 — Configure Response Latency	To do plan	AS	5
	SCRUM-41	US-09 — Test Mock Endpoints	To do plan	AS	8
	SCRUM-42	US-10 — Support Peak Request Load	In Progress	AS	5
	SCRUM-45	US-11 — Maintain Latency Accuracy	In Progress	AS	5
	SCRUM-44	US-12 — Reject Unsafe Specifications	To do plan	AS	11
	SCRUM-43	US-13 — (Unlabeled)	To do plan	AS	11

Epics and Their User Stories

Backlog Board

Spaces

OpenAPI Mock Server Generator  

 Summary  **Backlog**  Board  Docs  Development 

   Filter   

Epic 

 No epic

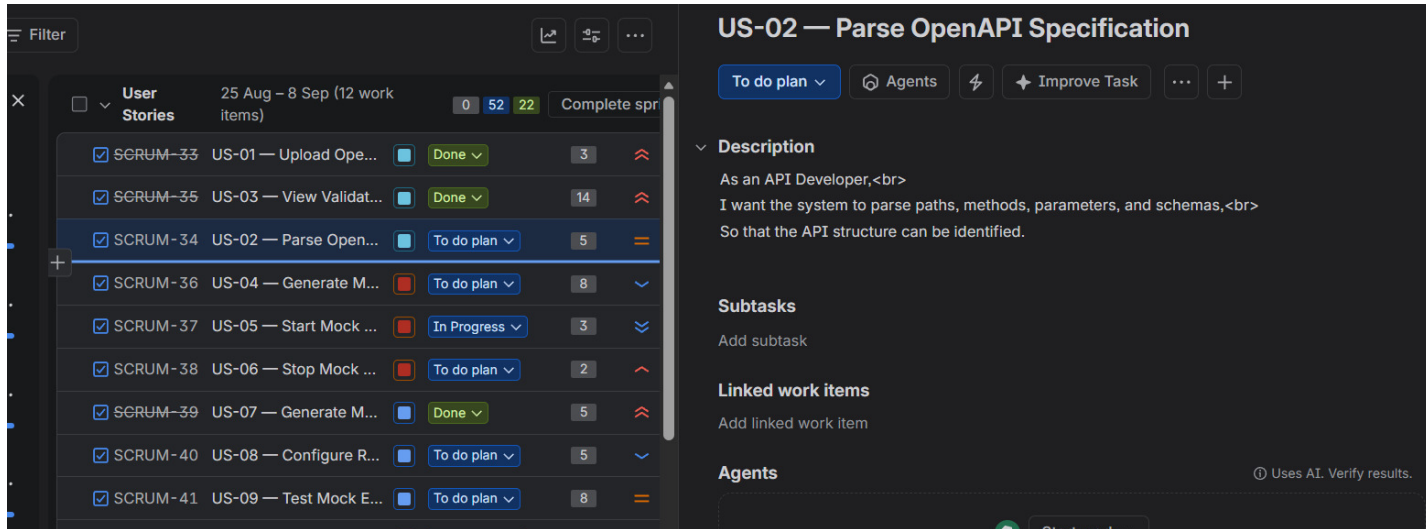
-  Epic 1: OpenAPI Specification Manager 
-  Epic 2: Mock Server Generation 
-  Epic 3: Response Simulation and Testing 
-  Epic 4: Performance, Security and Reliability 

☐  **User Stories** 25 Aug – 8 Sep (12 work items) 0 74 0 Complete sprint 

☒	SCRUM-33	US-01 — Upload OpenAPI Specification	 Epic 1: OpenAPI S...	To do plan 	3		
☒	SCRUM-34	US-02 — Parse OpenAPI Specification	 Epic 1: OpenAPI S...	To do plan 	5		
☒	SCRUM-35	US-03 — View Validation Errors	 Epic 1: OpenAPI S...	To do plan 	14		
☒	SCRUM-36	US-04 — Generate Mock Endpoints	 Epic 2: Mock Serv...	To do plan 	8		
☒	SCRUM-37	US-05 — Start Mock Server	 Epic 2: Mock Serv...	To do plan 	3		
☒	SCRUM-38	US-06 — Stop Mock Server	 Epic 2: Mock Serv...	To do plan 	2		
☒	SCRUM-39	US-07 — Generate Mock Responses	 Epic 3: Response ...	To do plan 	5		
☒	SCRUM-40	US-08 — Configure Response Latency	 Epic 3: Response ...	To do plan 	5		
☒	SCRUM-41	US-09 — Test Mock Endpoints	 Epic 3: Response ...	To do plan 	8		
☒	SCRUM-42	US-10 — Support Peak Request Load	 Epic 4: Performan...	To do plan 	5		
☒	SCRUM-45	US-11 — Maintain Latency Accuracy	 Epic 4: Performan...	To do plan 	5		

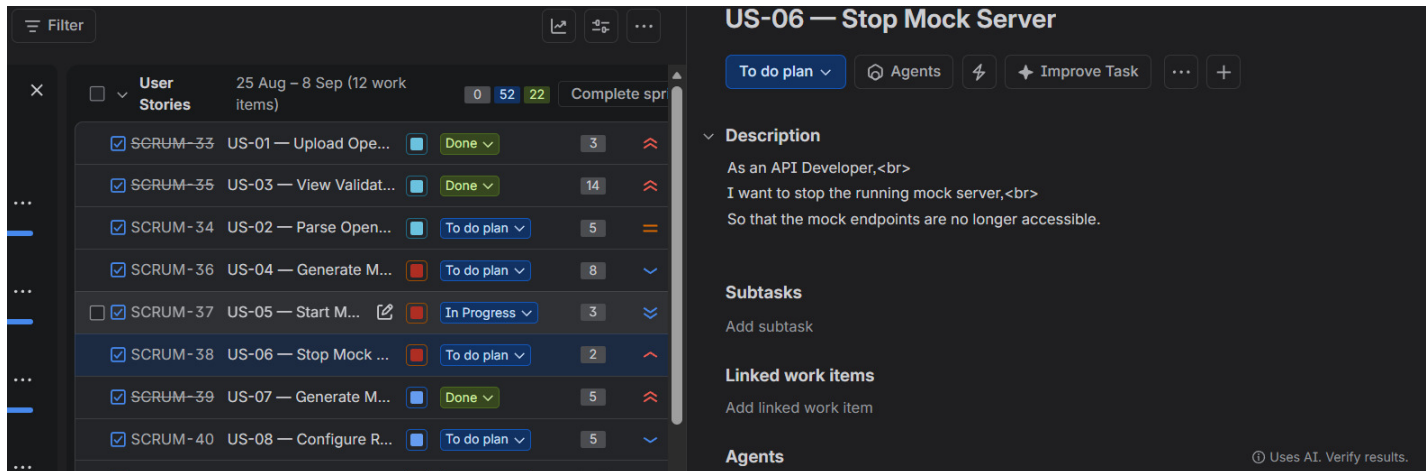
Jira Backlog Board

2. User Story Details



The screenshot shows the Jira interface. On the left, a list of user stories is displayed under the 'User Stories' filter. The stories are: SCRUM-33 (US-01 — Upload Ope...), SCRUM-35 (US-03 — View Validat...), SCRUM-34 (US-02 — Parse Open...), SCRUM-36 (US-04 — Generate M...), SCRUM-37 (US-05 — Start Mock ...), SCRUM-38 (US-06 — Stop Mock ...), SCRUM-39 (US-07 — Generate M...), SCRUM-40 (US-08 — Configure R...), and SCRUM-41 (US-09 — Test Mock E...). The story SCRUM-34 (US-02 — Parse Open...) is selected. On the right, the details for US-02 are shown. The title is 'US-02 — Parse OpenAPI Specification'. The status is 'To do plan'. The description is: 'As an API Developer,
I want the system to parse paths, methods, parameters, and schemas,
So that the API structure can be identified.' The subtasks section is empty. The linked work items section is empty. The agents section shows 'Uses AI. Verify results.'

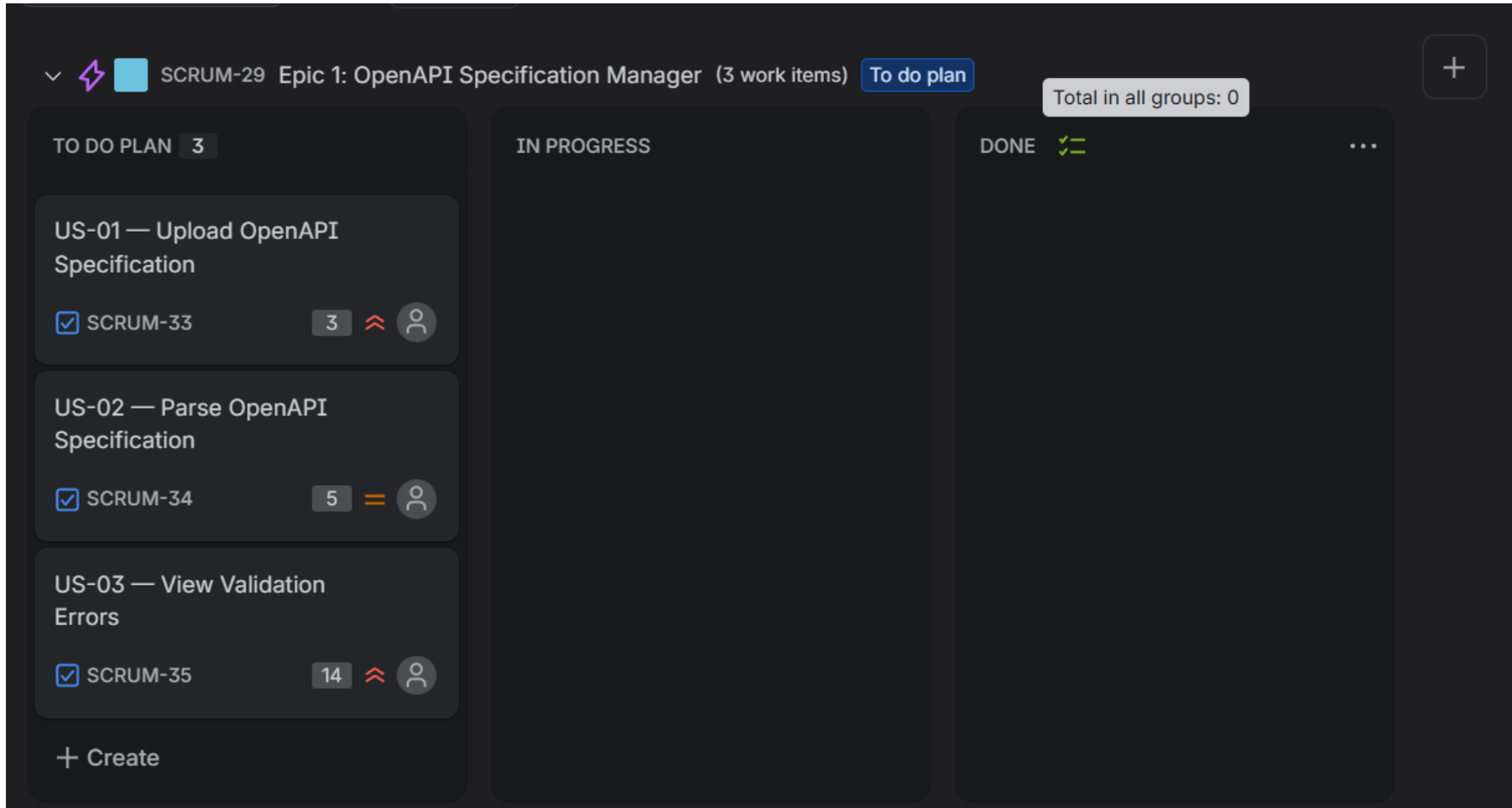
Epic Story 2 with Description



The screenshot shows the Jira interface. On the left, a list of user stories is displayed under the 'User Stories' filter. The stories are: SCRUM-33 (US-01 — Upload Ope...), SCRUM-35 (US-03 — View Validat...), SCRUM-34 (US-02 — Parse Open...), SCRUM-36 (US-04 — Generate M...), SCRUM-37 (US-05 — Start M...), SCRUM-38 (US-06 — Stop Mock ...), SCRUM-39 (US-07 — Generate M...), SCRUM-40 (US-08 — Configure R...), and SCRUM-41 (US-09 — Test Mock E...). The story SCRUM-38 (US-06 — Stop Mock ...) is selected. On the right, the details for US-06 are shown. The title is 'US-06 — Stop Mock Server'. The status is 'To do plan'. The description is: 'As an API Developer,
I want to stop the running mock server,
So that the mock endpoints are no longer accessible.' The subtasks section is empty. The linked work items section is empty. The agents section shows 'Uses AI. Verify results.'

Epic 2 — Story 6 with Description

3. Priorities and Story Points



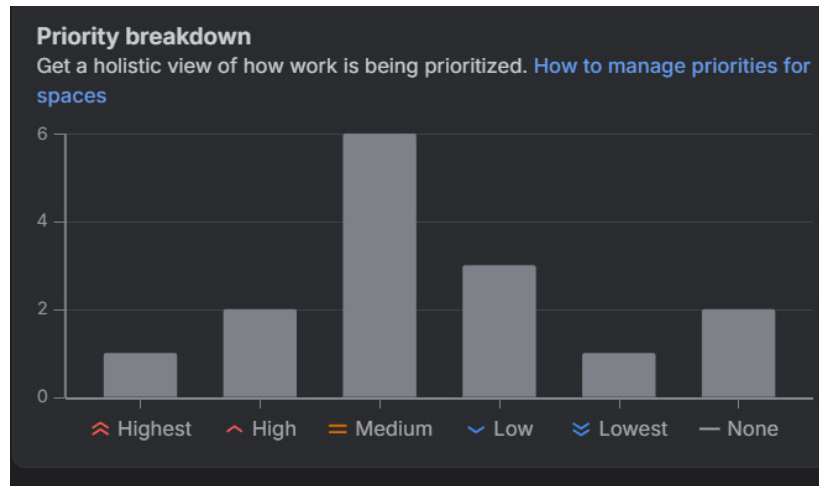
The screenshot displays a Jira Scrum board for the project "SCRUM-29 Epic 1: OpenAPI Specification Manager (3 work items)". The board is divided into three columns: "TO DO PLAN", "IN PROGRESS", and "DONE". The "TO DO PLAN" column contains three user stories, each with a checkbox, a story ID, a story point value, a priority indicator, and an assignee icon.

User Story	Story ID	Story Points	Priority	Assignee
US-01 — Upload OpenAPI Specification	SCRUM-33	3	High	[Assignee]
US-02 — Parse OpenAPI Specification	SCRUM-34	5	Medium	[Assignee]
US-03 — View Validation Errors	SCRUM-35	14	High	[Assignee]

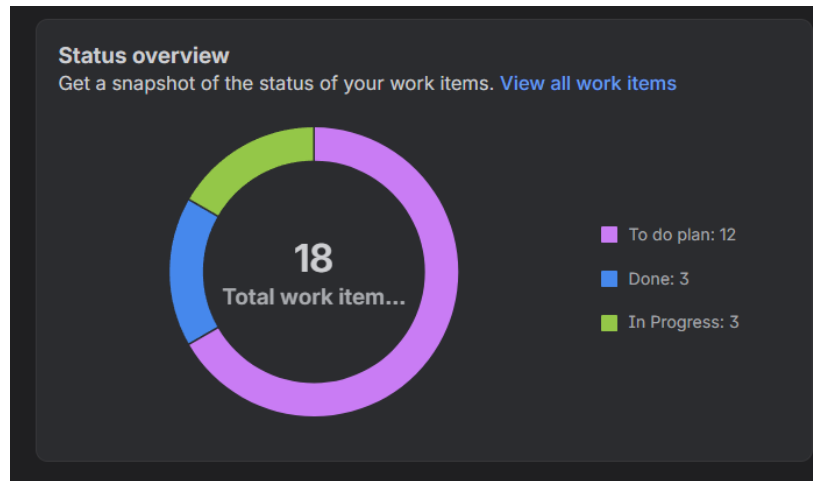
The "IN PROGRESS" and "DONE" columns are currently empty. A "Total in all groups: 0" label is visible above the "DONE" column. A "+ Create" button is located at the bottom of the "TO DO PLAN" column.

User Stories with Story Points and Priority

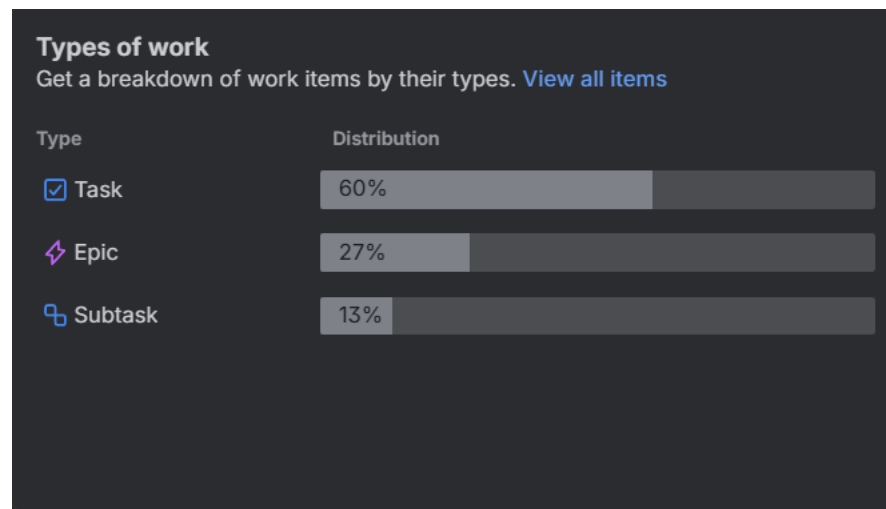
Backlog Work and Sprint Overview



Priority Breakdown



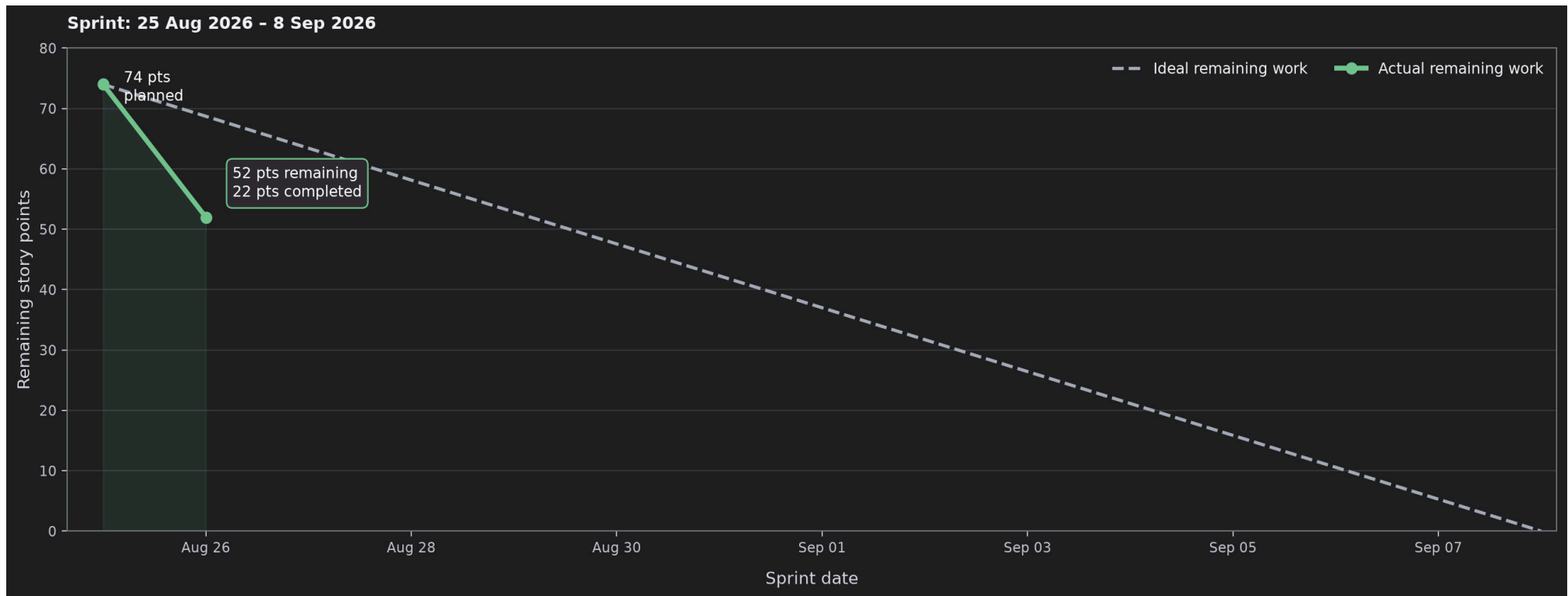
Status Overview



Work Types: Epics, Stories and Tasks in the Sprint Backlog

4. Sprint Analytics

The chart compares remaining story points with the ideal completion guideline across the sprint.



Sprint Burndown Chart

5. Reflection Questions

1. Did your estimations reflect the actual effort?

Mostly. The story points captured relative effort: simple actions received fewer points, while validation and security work received more. Since this was a simulated sprint, the estimates still need to be checked against real development time.

2. Was your backlog well-prioritized?

Yes. The core workflow came first: uploading and parsing the OpenAPI file, generating endpoints, and running the mock server. Testing, performance, and security followed. Future planning should also balance priority with available capacity.

3. How did your simulated sprint align with your plan?

Partially. The sprint completed 22 of 74 story points (about 30%), with 3 items done, 3 in progress, and 12 still planned. This indicates that the selected sprint scope was larger than the simulated capacity.

4. What insights did the burndown chart give about your team's capacity?

The chart showed an early drop from 74 to 52 remaining points, but a large amount of work was still unfinished. The team should commit to fewer high-value stories and review progress daily to avoid carrying excessive work forward.